

TECHNICAL SKILLS

Artificial Intelligence & Machine Learning: PyTorch, TensorFlow, OpenCV, YOLOv8, Core ML, Scikit-learn

Full Stack & Software Engineering: React, Next.js, Angular, TypeScript, JavaScript, HTML/CSS, Tailwind, Spring Boot, Node.js, Grails, Groovy, SQL (Oracle, MySQL)

Tools & Infrastructure: Python, C++, Java, Bash, Git, Docker, Linux/Ubuntu, GPU/CUDA, Postman, Jira, Bitbucket

EDUCATION

University of Sheffield

Sheffield, UK

*M.S. in Advanced Computer Science, **With Merit***

Sep 2024 – Oct 2025

- **Dissertation:** Multi-Target Speaker Extraction for Conversational Enhancement – deep learning with PyTorch (CHiME-9 ECHI Challenge)
- **Relevant:** Parallel Computing with GPUs, Speech Processing, Software Reengineering, Speech Technology, Mobile Development, OOP & Software Design

Universidad Peruana de Ciencias Aplicadas (UPC)

Lima, PE

B.S. in Computer Science, GPA: 3.5/4.0

Jan 2018 – Jul 2023

- **Concentrations:** Machine Learning & Computer Vision
- **Dissertation:** DriveSafe – ML-powered iOS drowsiness detection using PyTorch, Core ML, and YOLOv8

WORK EXPERIENCE

Mandü HR

Lima, PE

AI Engineer

Aug 2026 – Present

- Built a Python pipeline normalizing 70k+ job titles into occupation families and subfamilies, combining domain embeddings, LLMs, and fuzzy matching to generate taxonomy nodes
- Exposed the taxonomy through a FastAPI service powering search and classification over HR data
- Shipped an Angular mini front-end to browse and validate families, subfamilies, and roles

Bevertec

Lima, PE

Fullstack Developer

Sep 2022 – Sep 2023

- Built RESTful APIs in Spring Boot for a platform with 10,000+ daily users, including endpoints for user permissions and data retrieval
- Rewrote slow Oracle SQL queries behind internal reports, cutting load time from 8s to 4s
- Built MVC applications with Groovy and Grails as part of 2-week Agile sprints alongside design and QA
- Ran code reviews, managed releases through Bitbucket, and tested API responses and error handling in Postman

Rebels 247

Lima, PE

Full Stack Web Developer

Nov 2021 – Feb 2022

- Built websites with React and Tailwind CSS, deployed on Vercel with automated builds
- Created a library of reusable React components (buttons, forms, modals) reused across 5+ client projects, cutting development time by roughly 40%
- Integrated the ActiveCampaign API for email campaigns and added Google Analytics and Facebook Pixel for activity tracking

INET

Lima, PE

Front End Developer

Oct 2021 – Nov 2021

- Developed Angular services in TypeScript to connect to REST APIs for user input data, using HTTP interceptors to attach authentication tokens
- Built a responsive site in SASS with a mobile-first approach and basic SEO practices
- Met weekly with the client to review and redesign components based on feedback

TECHNICAL PROJECTS

- **Multi-Target Speaker Extraction (M.Sc. Dissertation):** Built a deep learning system for conversational speech enhancement in PyTorch as part of the CHiME-9 ECHI Challenge, separating multiple target speakers from noisy multi-speaker audio
- **WRO 2024 – Autonomous Driving Robot:** Programmed an Ackermann-steered robot for autonomous circuit navigation and obstacle avoidance using computer vision, 2D LiDAR, and PID control – 2nd Place at WRO Qualifiers (Peru), 3rd Place at the Open Challenge (Costa Rica)
- **DriveSafe – Driver Drowsiness Detection (iOS):** ML-powered iOS app for driver drowsiness detection (B.Sc. dissertation), built with PyTorch, Core ML, and YOLOv8 on a custom-trained dataset, running on-device at 68 fps
- **YOLO-MD – Face Mask Detection:** Real-time face mask detection with YOLOv5, containerized with Docker and deployed on Heroku for COVID-era compliance monitoring

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Machine Learning with TensorFlow on Google Cloud:** Google Cloud
- **Learn SQL Basics for Data Science Specialization:** University of California, Davis
- **Foundations of User Experience (UX) Design:** Google
- **Scrum Fundamentals Certified (SFC):** SCRUMstudy
- **EF SET English Certificate:** 82/100 (C2 Proficient) – EF Standard English Test

LEADERSHIP & INVOLVEMENT

Marsworks – Scarab Rover Project

Sheffield, UK

Software Team Lead

Oct 2024 – Oct 2025

- Designed the autonomous navigation pipeline for a Mars rover analog using ROS2, RTAB-Map SLAM, and Nav2, achieving point-to-point autonomous travel and 3D mapping of the environment
- Led a team of 6 on ROS2 development, sensor integration (RealSense D435i depth camera), and navigation algorithms, working alongside the mechanical and electrical sub-teams
- Built the MAVROS-ArduPilot communication layer handling real-time control and telemetry for the rover
- Tested navigation behaviours in Gazebo and Isaac Sim before deployment, catching and fixing control issues early

Robotek

Lima, PE

Robotics Coach & Mentor

2023 – 2024

- Coached two teenage students in computer vision, PID control, Arduino, and Raspberry Pi for autonomous robotics
- Led team to 2nd Place at World Robot Olympiad (Peru) and 3rd Place at WRO Open Championship (Puerto Rico), competing against teams from USA, Canada, China, and Korea
- Prepared team for WRO World Finals (Turkey) with upgraded hardware incorporating ROS2 and LIDAR sensors

BLK Robotics

Lima, PE

Co-Founder & Competitor

2023 – 2024

- Co-founded robotics club and competed in national competitions securing 1st Place (Bipedal Robot Race) and 2nd Place (Mini Sumo) at RoboJam 2024